Results: Bandit-Based Material Screening

Learning curves and algorithm comparison | Source: executed notebook (results.json and figures)

Setup. 60 real steels sampled from the matminer *steel_strength* dataset (yield strength, MPa). Reported strengths are used as hidden true means; measurement noise is simulated (Gaussian, SD 100 MPa). Budget 300 tests, 30 independent seeds, same pool and noise for every algorithm. Pool mean 1,409 MPa, best steel 2,487 MPa. An oracle that always tested the best steel would collect about 746,190.

Summary table

Algorithm	Cumulative reward (MPa, summed)	Cumulative regret (MPa, summed)	Final simple regret (MPa)	Best-steel ID (%)	Median tests to first top-10% steel	Top-10% share, last 100 tests (%)
Random	422,827 ± 2007	323,156 ± 1824	0.0 ± 0.0	100	7	10.3
Epsilon-Greedy	656,257 ± 1562	89,898 ± 1401	0.0 ± 0.0	100	8	91.4
UCB1	434,094 ± 652	312,026 ± 227	0.0 ± 0.0	100	22	13.7
Thompson Sampling	685,205 ± 13991	60,954 ± 13795	22.0 ± 30.0	93	6	96.8
UCB1 (c=0.1) (extra)	673,483 ± 860	72,594 ± 504	0.0 ± 0.0	100	22	98.3

Mean over 30 seeds, plus-minus values are 95% confidence intervals across seeds. Reward is observed (noisy); regret uses the true underlying strengths.

Interpretation

- **Thompson Sampling** earned the most reward (685,205) and had the lowest regret (60,954). It concentrated 96.8% of its last 100 tests on top-10% steels. It is also the only method that sometimes missed the best steel (93% identification), because it tested only about 39 of 60 steels.
- **UCB1 with c = 0.1** (extra variant) and **Epsilon-Greedy** were close behind (673,483 and 656,257) and found the best steel in every run. UCB1 (c = 0.1) was within the uncertainty of Thompson Sampling's reward and much more consistent across seeds.
- **Standard UCB1 (c = 1)** stayed close to Random (434,094 vs 422,827). With 60 steels and 300 tests its exploration bonus is still large, so it keeps exploring. Its result depends strongly on the exploration scale.
- Paired against Random (same noise), the reward gain was 262,378 for Thompson Sampling, 250,656 for UCB1 (c = 0.1), 233,430 for Epsilon-Greedy and 11,267 for standard UCB1.
- **Identification does not separate the methods here.** The best steel is far above the rest of the pool, so even Random identified it in 100% of runs. Reward and regret are the informative metrics.
- Caveats: noise is simulated and assumed known to Thompson Sampling, one pool seed and budget were tested, and Thompson Sampling's prior (mean 1000 MPa) is below the pool mean.

Learning curves and plots

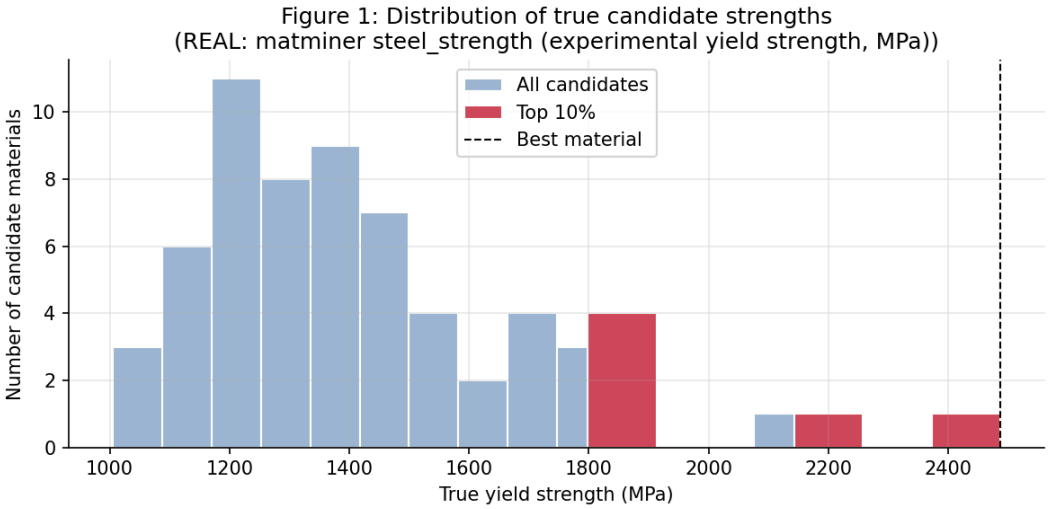


Figure 1: Distribution of true candidate strengths (real matminer sample).

The pool is concentrated around 1,409 MPa. The best steel (2,487 MPa) stands well above the rest, and the top 10% start at 1,798 MPa.

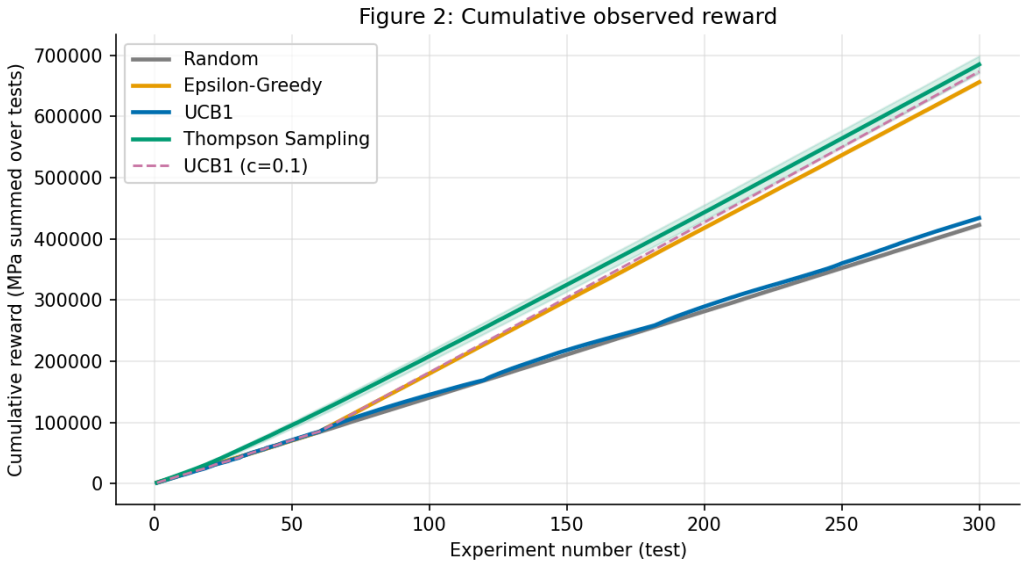


Figure 2: Cumulative observed reward (mean and 95% CI over seeds).

Thompson Sampling, UCB1 (c = 0.1) and Epsilon-Greedy pull away from Random after the first pass over the steels. Standard UCB1 and Random almost overlap.

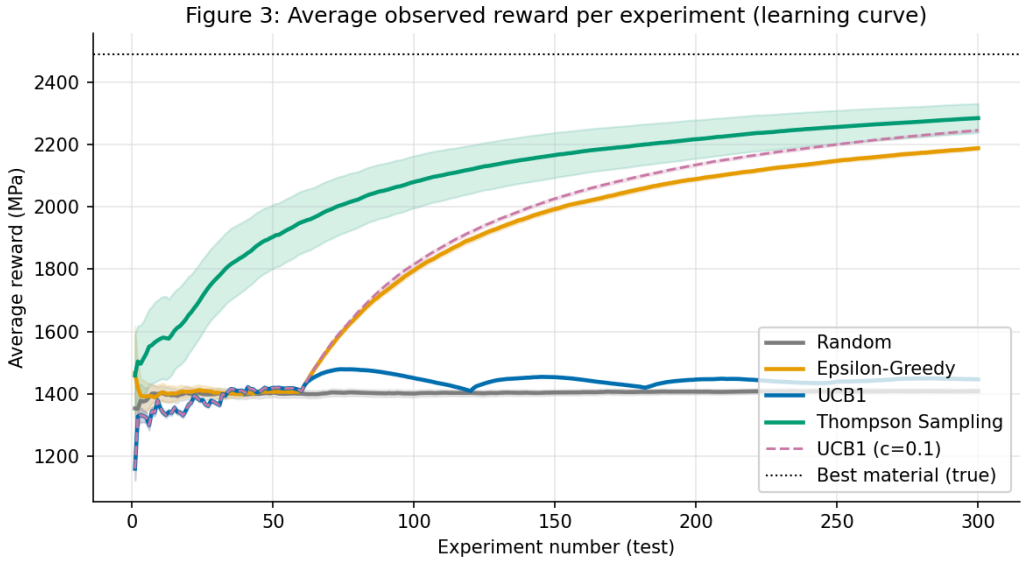


Figure 3: Average observed reward per experiment (learning curves).

Thompson Sampling learns fastest. Epsilon-Greedy and UCB1 (c = 0.1) start improving after each steel has been tested once. Random stays near the pool mean; the dotted line is the best steel.

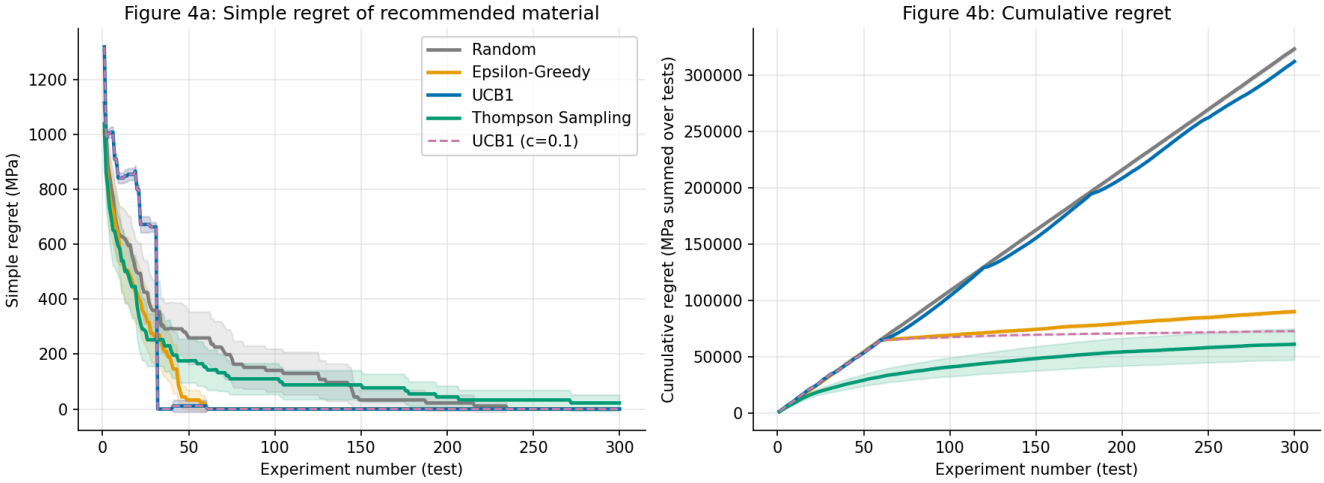


Figure 4: Simple regret of the recommended steel (a) and cumulative regret (b).

Simple regret reaches 0 for every method except Thompson Sampling, which ends at 22.0 ± 30.0 MPa because it sometimes recommends the second-best steel. Cumulative regret is lowest for Thompson Sampling.

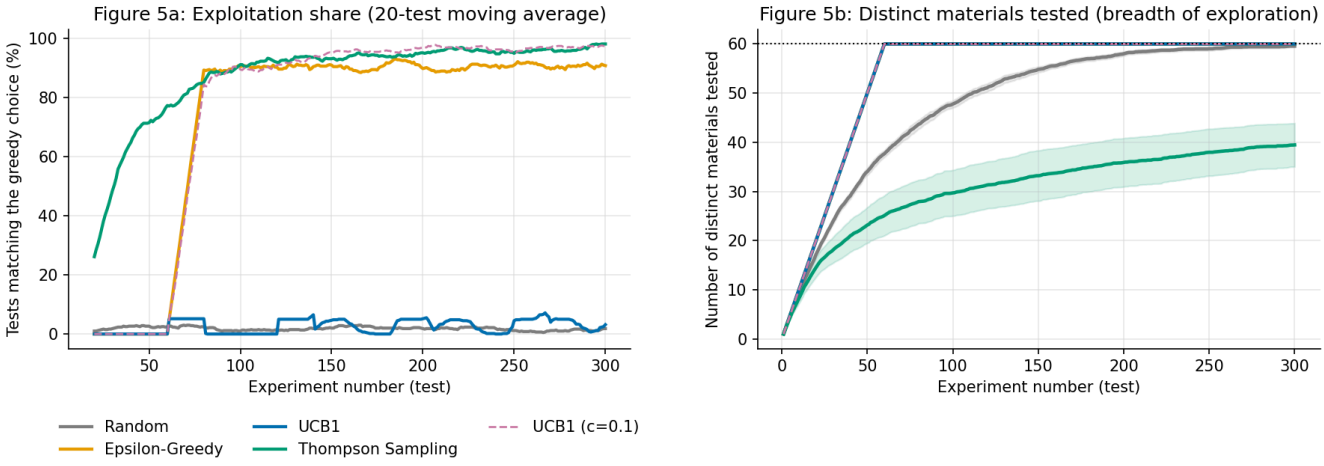


Figure 5: Exploitation share (a) and number of distinct steels tested (b).

Greedy-choice share: Thompson Sampling 86.6%, UCB1 (c = 0.1) 75.7%, Epsilon-Greedy 72.6%, standard UCB1 2.6%, Random 1.8%. Thompson Sampling tests only about 39 of 60 steels; the others test almost all.

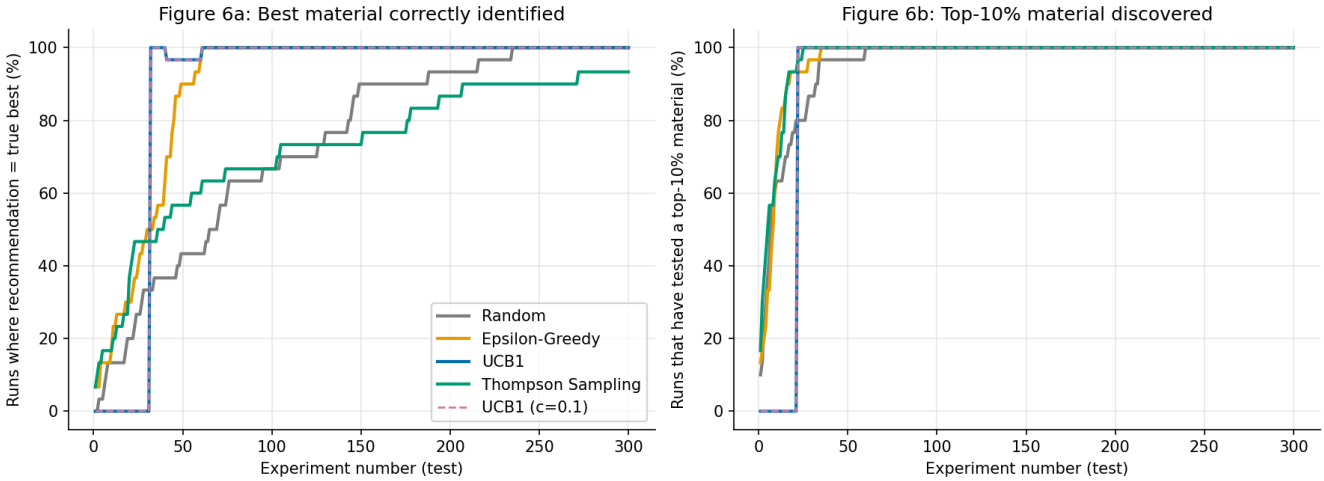


Figure 6: Best-steel identification (a) and discovery of a top-10% steel (b).

All methods find a top-10% steel early. Median first-hit test: 7 (Random), 6 (Thompson Sampling), 8 (Epsilon-Greedy), 22 (both UCB1 variants). UCB1 tests steels in list order first, so its value mostly reflects pool order.

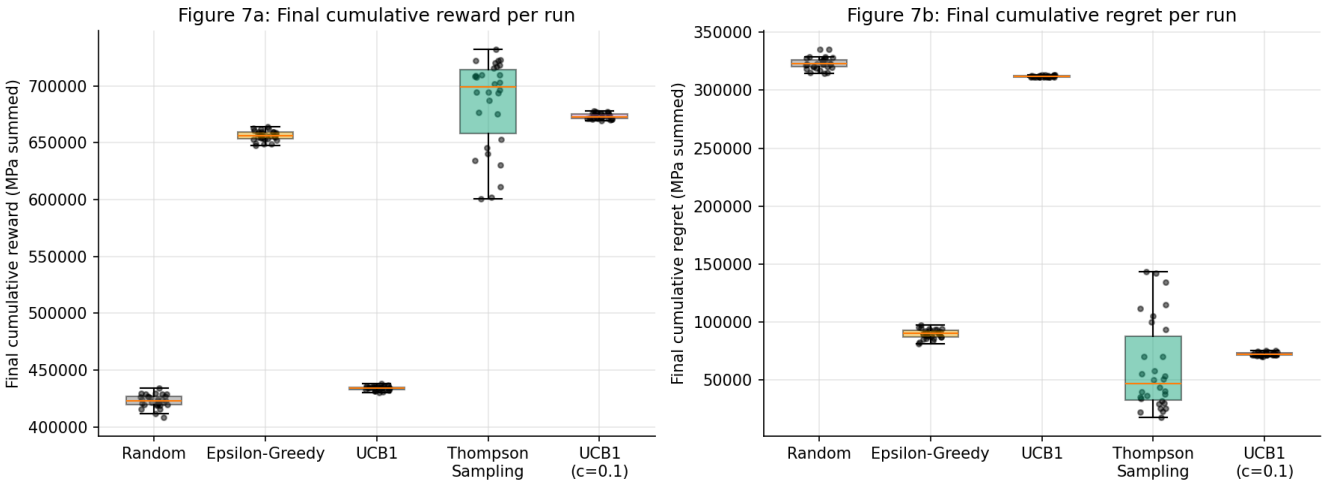


Figure 7: Final cumulative reward and regret across the independent runs.

Thompson Sampling has the highest reward but also the widest spread across seeds (95% CI of about 13,991 MPa). UCB1 (c = 0.1) is the most consistent.

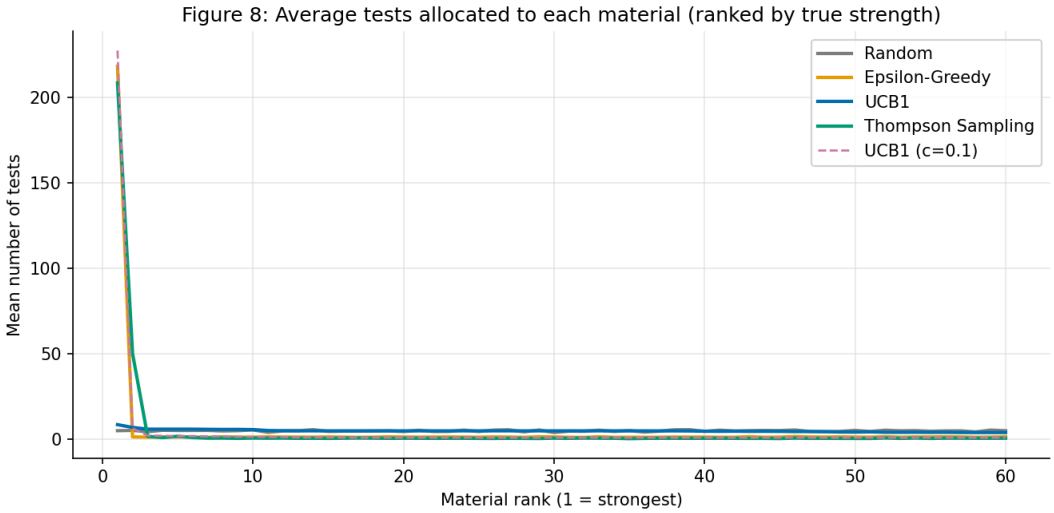


Figure 8: Average number of tests given to each steel, ranked by true strength.

Thompson Sampling, Epsilon-Greedy and UCB1 (c = 0.1) put most of their 300 tests on the single strongest steel (roughly 200 or more per run on average) and very few on the rest. Random and standard UCB1 spread tests almost evenly.